Personal info

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EDUCATION

University of Bristol | MSc Robotics (In Progress) | 2025.09-2026.11 (Expected, course ends in May)

Core modules: Artificial intelligence for Robotics, Machine Vision, Human-Robot Interaction, Robot Learning for Control, Robotics Science & Systems

Guangdong Neusoft Institute – University of West of England(Joint Programme) | BEng Software Engineering | GPA 3.66/5 (Ranked 1 st in major) | 2021.09–2025.06

Core modules: Object-Oriented Programming, Software Design & Analysis, Advanced Algorithms, Computer Networks, Data Structures, Software Testing

PROJECTS

Multimodal Vision-Language Aesthetic Analysis Agent System

Individual Final Year Project | Sep 2024 – Feb 2026

- Built a **vision-language multimodal system (VLM)** for image aesthetic scoring and visual analysis generation
- Used **Swin Transformer** for visual feature extraction and BLIP for image captioning; designed a **Mixture-of-Experts (MoE**, 6 experts + gating) for adaptive fusion of visual and textual features
- Achieved SRCC 0.6806 on TAD66K dataset, improving +169% over vision-only and +523% over text-only models
- Developed a **RAG-based** aesthetic knowledge module, retrieving photography rules (composition, lighting, color) to enhance interpretability
- Packaged the system as a multimodal image analysis agent, automatically orchestrating scoring, captioning, and report generation
- Deployed a **Flask web demo** supporting image upload → scoring → auto-generated analysis report, with ~1.9s inference per image

Lightweight Apple Counting with Cross-Scenario Generalisation

Machine Vision Course Project (Team of 5) | Oct 2025 – Dec 2025

- Built a **regression-based CNN model** for apple counting under occlusion and illumination variations, achieving 87.06% accuracy
- Developed a lightweight model based on EfficientNet-B0, integrating **LBP texture features + Channel Gate + CBAM attention** for RGB-LBP 4-channel fusion
- Achieved MAE 0.169 / ACC 87.06% on MinneApple dataset, reducing MAE by ~19.5% vs baseline
- Designed **ablation studies** (Baseline / Gate / CBAM / Full) to validate the synergy of attention and channel gating mechanisms
- Conducted **cross-domain generalisation experiments** on MinneApple / Combined / Natural datasets, analysing performance under domain shift
- Performed **failure case analysis using Grad-CAM** + error distribution, identifying challenges in occlusion, specular reflection, and dense clusters
- Implemented **ONNX Runtime** inference with **INT8 static quantisation**, reducing latency compared to FP32

SKILLS

Data & Algorithms: Proficient in SQL and relational databases; Strong experience in data mining, feature engineering, data cleaning, and optimisation (Pandas, NumPy)

Machine Learning & Deep Learning: Proficient in PyTorch; Familiar with CNN, Transformer, Attention mechanisms, CBAM, and Mixture-of-Experts (MoE); Understanding of multi-tier system design for ML systems

Engineering & Deployment: Linux environment, Git version control, Docker containerisation; Strong capability in system optimisation and complexity analysis

Programming: Python, Java, C++; Solid foundation in Object-Oriented Programming and Data Structures

Software Development: Familiar with multi-tier architecture and RESTful API design; Full-stack development with Flask, Django, Vue3

AWARDS

- Outstanding Undergraduate Thesis Award (Top 1 in cohort), Dec 2024
- Second Prize, National E-commerce Innovation Competition, Jun 2024
- Bronze Award, "Challenge Cup" Guangdong Academic Competition, May 2023